

Asset Analysis: 123

Asset ID:	3	Assessment Date:	2026-07-13
Location:	123	Status:	APPROVED & REVIEWED

1. Executive Summary

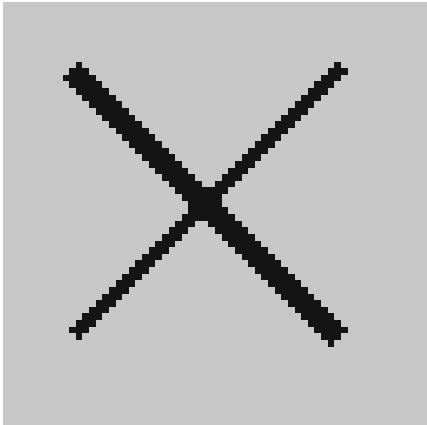
The asset, designated as "123", has been evaluated and determined to pose a medium risk level based on the diagnostic results. The overall health status of the asset is compromised due to detected cracks/scratches, despite exhibiting normal machine sound. The risk assessment is primarily driven by anomalies in temperature, vibration, and pressure readings.

2. Diagnostic Predictions

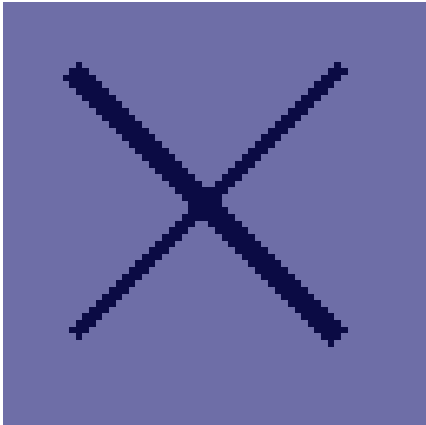
Diagnostic Module	Initial AI Prediction	Final Adjusted Status
Tabular Time-Series (LSTM)	Medium Risk	High Risk
Visual Damage (CNN)	Damaged (Cracks/Scratches)	Damaged (Cracks/Scratches)
Sound Analysis (ANN)	Normal Noise	Anomalous Sound

3. Explainable AI (XAI) Insights

Time-Series Perturbation Saliency: Temperature: 40.6% impact • Vibration: 32.3% impact • Pressure: 27.2% impact



Original Image



Visual Grad-CAM Saliency Highlight

4. Technical Risk Reasoning

The technical analysis reveals that the asset's risk level is influenced by a combination of factors, including time-series data indicating a medium risk level as predicted by the LSTM model. The feature importance analysis highlights that temperature, vibration, and pressure anomalies are the primary contributors to this risk assessment, with temperature being the most significant factor at 40.57%. Furthermore, the CNN-based visual inspection has detected cracks/scratches with a confidence level of 0.50, which corroborates the medium risk level. In contrast, the machine sound diagnostic using ANN indicates normal operation, suggesting that the asset's functional integrity is currently intact despite the physical damage.

5. Recommended Mitigation & Compliance Steps

- To mitigate the identified risks and ensure compliance with regulatory guidelines, the following action steps are proposed:
- Conduct a thorough maintenance check to address the detected cracks/scratches, which may involve repairing or replacing damaged components to prevent further deterioration.
- Implement real-time monitoring of temperature, vibration, and pressure levels to enable prompt detection of anomalies and allow for swift corrective action, thereby reducing the risk of asset failure.
- Develop and implement a regular inspection schedule that incorporates both visual assessments using CNN-based tools and audio diagnostics using ANN-based methods to ensure the early detection of potential issues, facilitating proactive maintenance and minimizing downtime.

6. Human-in-the-Loop Analyst Verification

Analyst Remarks: jhfdslhdsljhf

Reviewed By: lhdslfhsljdf

Review Date: 2026-07-13